

MIE230 More Interesting Problems

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Note: Problems 1-8 cover the midterm material. Anything beyond is in the final. Problems get progressively harder as you go along. The starred problems are particularly challenging, especially problem 8.

Problem 1: Electrons Go BOOOM!

Let Z be a random variable over the complex plane, $\mathbb{C}$. This is the sample space of all possible spins of an electron. The spin is represented by the following spinor field:

$$\sigma = \left[A \det \begin{pmatrix} x & 0 \\ 0 & -y^2 \end{pmatrix}, B \left\{ 3 - i \cdot \det \begin{pmatrix} e^{-x^2} & -1 \\ 1 & e^{-y^2} \end{pmatrix} \right\} \right]$$

These matrices are called Pauli matrices. Sometimes, the sum of probabilities does not add up to 1 in which case, we have to introduce a Normalization term to scale things down. That is, in fact, the roles of A and B . Determine the values of A and B which normalize this distribution if the CDF of the electron is given by:

$$\psi(Z \leq z) = \iint_{(-\infty, -\infty)}^z \sqrt{|\sigma|} dA$$

Where $|\sigma|$ represents the magnitude of the spinor field.

Problem 2: Mechanics of Curves

Let $\mathcal{M}$ be the surface described by the set

$$\mathcal{M} = \{(x, y, z) \in \mathbb{R}^3 \mid z = x^2 + y^2\}$$

We have a curve $\mathbf{r}_1 \in \mathcal{M}$ that is a helix that wraps around our surface $\mathcal{M}$ with a speed vector of

1. Suppose that such curve represents the trajectory of a particle. Find the equation of this curve. (Hint: parametrize!) Suppose that this trajectory is due to the particle being in centripetal/helical motion under a varying centripetal force. Suppose that the particle starts to follow a straight line path at time $t = 12s$. Determine the basis vectors most natural for analyzing trajectory of the particle after $t = 12s$. Calculate the total length the particle traveled in the helical trajectory.

Problem 3: Gaussian Integrals

An important, and classic integral in the theory of statistics and Quantum Mechanics is of the form:

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} dx, \alpha \in \mathbb{R}$$

Determine the value of this integral as a function of α .

Problem 4: Funky Volumes

Consider the following helix:

$$\mathbf{r}(t) = [2 \sin t, 2 \cos t, 3t], \quad t \in [0, 4\pi]$$

In CAD software such as SolidWorks, there is an option called "sweep". This allows you to make a "pipe" shape with a circular cross-section that follows the path of a curve. For a cross-section of diameter, $D = 3$ units, what is the volume of the solid created ?

Problem 5: Compressible Flow Curve

Consider an incompressible fluid where $\mathbf{v} = u(x, y, z)\mathbf{i} + v(x, y, z)\mathbf{j} + w(x, y, z)\mathbf{k}$ is the velocity field. Consider this fluid flowing in the path of a curve

$$\mathbf{r}(t) = [e^{-t}, e^{-t}, -t - 20]$$

Calculate the components of the normal and tangential acceleration of the fluid at some point (x_0, y_0, z_0) given that:

$$u = x^2y^3 + z^4 \quad v = \sin(x) - \cos(y)\cos(z) \quad w = xyz - e^{x+y+z}$$

Problem 6: The Curvey Curves!

Consider the following parametric curves in $\mathbb{R}^3$:

$$\mathbf{r}_1(t) = [\cos t, 2 \tan t, \cos t]$$

$$\mathbf{r}_2(t) = [\sinh t, 2 \cos t, \sin t]$$

$$\mathbf{r}_3(t) = [\sinh t, 3, \sin t]$$

Firstly, determine the interval over which the intersection of these curves forms a surface given that the parameter $t \in \mathbb{R}$. Then, approximate the volume between this surface and the surface $f(r, \theta) = r - 2$ for $t \in [-\pi/2, \pi/2]$ (Hint: you can approximate solutions of equations and integrals with a certain series that rhymes with Taylor S..ies).

*Problem 7: Topology of Curve Projections

Use all information you need from problem 2. Suppose we shine a light on our curve in the direction $[1, 1, -1]$. Calculate the arc length of the projection of this curve on the plane $z = x - y - 1$.

*Problem 8: The Hairy Ball Theorem

Yes, this is the name of a real theorem. Consider the unit sphere:

$$S^2 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$$

Suppose we have a vector field, $\mathbf{F}(x, y, z) = F_1(x, y, z)\mathbf{i} + F_2(x, y, z)\mathbf{j} + F_3(x, y, z)\mathbf{k}$ on this surface such that $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$. Show that there exists no continuous vector field that is always tangent to our sphere, S^2 .

Problem 9: Elliptic Ellipse

Consider the curve $\gamma = \{(x, y) \in \mathbb{R}^2 \mid x^2/a^2 + y^2/b^2 = 1\}$. Then calculate:

$$\int_{\gamma} (y \cos x) dx + (x + \sin x) dy$$

Problem 10: Aerodynamics of Cylinders

The following vector field describes the air velocity around a cylinder:

$$\mathbf{v} = U \left(1 - \frac{a^2}{r^2} \right) \cos \theta \mathbf{r} - U \left(1 + \frac{a^2}{r^2} \right) \sin \theta + \frac{\Gamma}{2\pi r} \boldsymbol{\theta}$$

Where $r, a, U > 0$ and $\Gamma \in \mathbb{R}$. Furthermore, let C be a circle of radius:

$$r(\theta) = a(1 + \varepsilon f(\theta)), \quad \theta \in [0, 2\pi]$$

Where $\varepsilon \in \mathbb{R}$ and $f(\theta)$ is a periodic function of period 2π . This function is chosen accordingly to make $r(\theta) > 0$.

Part 1: Do Your Integration Homework!

The circulation term Γ_C is defined as:

$$\Gamma_C = \oint_C \mathbf{v} \cdot d\mathbf{r}$$

Calculate this term and explain why the terms contributing to uniform flow integrate to 0. Show that $\Gamma_C = \Gamma$ for any smooth $f(\theta)$.

Part 2: The Grass Is Very Green

Calculate the scalar vorticity of the field, $\omega_z = (\nabla \times \mathbf{v}) \cdot \mathbf{k}$. Suppose D is the region enclosed by C , then calculate the circulation in using a different method than in part 1 OR explain why such task is not possible and introduce any correction terms necessary in order to make this possible.

Part 3: Taylor, Taylor, Taylor, ...

The field

$$\psi = U \left(1 - \frac{a^2}{r^2} \right) \sin \theta + \frac{\Gamma}{2\pi} \ln r$$

is called the stream function of $\mathbf{v}$. Show that $v_r = \partial\psi/r\partial\theta$ and vice versa for $v_\theta = -\partial\psi/\partial r$.

The impermeability condition on a solid boundary is that the normal velocity vanish. Expand this condition to first order for the ε term in $r(\theta)$. Derive a linear equation for $f(\theta)$ explicitly that enforces impermeability.

Part 4: Pressure Coefficients and Lift

Consider Bernoulli's equation: $p + \frac{1}{2}\rho|\mathbf{v}|^2 = p_\infty + \frac{1}{2}\rho U^2$ for steady, incompressible, irrotational flow. Using Bernoulli's equation, you find that the pressure coefficients are defined as:

$$C_p(\theta) = 1 - \frac{|\mathbf{v}|^2}{U^2}$$

Expand this term in terms of $f(\theta)$ and ε from the last part!

The total force on the body due to pressure is calculated by:

$$\mathbf{F} = - \oint_C p \mathbf{n} \cdot d\mathbf{r}$$

Derive the lift as a function of θ and expand to first order. How is this related to the circulation, Γ and free stream velocity, U ? Explicitly calculate the first two terms of the lift function. (Hint: You need to expand line elements, $d\mathbf{r}$ and $\mathbf{n}$ in terms of $C_p(\theta)$ and then calculate $\mathbf{F}$)

*Problem 11: Geodesics Over Funky Surfaces

It is 2028 and Elon is putting everybody on mars. They have also informed everyone that mars is not a sphere. In fact it is a surface,

$$\mathcal{M} = \{(x, y, z) \in \mathbb{R}^3 \mid z = m(t) \cos(n(t)x) + n(t) \sin(m(t)y), m(t) : \mathbb{R} \rightarrow \mathbb{R}, n(t) : \mathbb{R} \rightarrow \mathbb{R}\}$$

Elon has found out that this surface changes over time with respect to twice differentiable functions $m(t)$ and $n(t)$ due to cosmic rays and human activities on Mars. Your favorite Aunt, who moved to mars way before you that you really miss and want to visit every weekend, has bought a house at point $X = (x_0, y_0, z_0)$ on mars. Your house, is on point $Y = (x_1, y_2, z_2)$ and you do not have much money left to spend on transportation since you spent well over \$100K on your ticket to mars. Determine the minimum distance you would need to walk in order to get to your aunt's house.

*Problem 12: Gotta Know Your ODEs

You have become an aerodynamics for the Scuderia Ferrari F1 Team. Your job is to design the most hyper-futuristic airfoils for the rear wing of the car. Being the innovative engineer you are, you decide that your airfoil's boundary functions (the upper and lower surface) must satisfy the following condition:

$$y''(x) + 2y'(x) + 5y(x) = 0$$

The upper surface must also satisfy $y_U(0) = 0, y'_U(0) = 0.8$ and the lower surface satisfies $y_L(0) = 0$ and $y'_L(0) = -0.4$.

In CFD simulations, your results suggest that the pressure distribution on the upper surface obeys the following differential equation:

$$p''_U(y) + 2.4p'_U(y) + 2.44p_U(y) = 0$$

such that $p_U(0) = 0$ and $p'_U(0) = 4$. The pressure distribution on the lower surface follows a different equation:

$$p''_L(x) + 4p'_L(x) + 5p_L(x) = 0$$

where $p_L(0) = 0, p'_L(0) = -1$. These are both for a symmetric, 0 AoA airfoil.

Part 1: Les Functions ?

Determine the functions for lower and upper surface pressure distributions and the airfoil profile. The first point on the airfoil which touches the air is known as the leading edge, and the last point is called the trailing edge. What is the distance from the leading edge to the trailing edge ? Calculate the radius of curvature for the airfoil surface profiles along its length.

Use these links to check your answer and visualize the other parts:

Airfoil and Boundary: <https://www.desmos.com/calculator/60fihljnuy>

Pressure and Respective Velocity Distributions: <https://www.desmos.com/calculator/hkpak4lrh4>

Part 2: Lift and Drag Functions

The lift and drag generated by an airfoil due to the flow field is given by the projection of the pressure normal vectors onto the **i** and **j** directions respectively. For continuous pressure, it would be:

$$L = \oint_Y p \mathbf{n} \cdot \mathbf{i} \, d\mathbf{r} \quad D = \oint_Y p \mathbf{n} \cdot \mathbf{j} \, d\mathbf{r}$$

Where Y is the contour formed by the two curves $y_U(x)$ and $y_L(x)$. Calculate the total lift and drag generated by the airfoil at 0° AoA as a function of velocity $\mathbf{v} = [v_1, v_2]$.

Part 3: Circulation and Velocity

Recall from question 10 that the circulation of a vector field is given by:

$$\Gamma = \oint_C \mathbf{v} \cdot d\mathbf{r}$$

Calculate the net circulation for a cardioid region C . As a courtesy (yeah I know, I'm very generous), here is the cartesian equation:

$$(x^2 + y^2 - x)^2 = 8((x + 2)^2 + y^2)$$

take a velocity field of $\mathbf{v} = [s^2, -2e^{-s} \sin t - 3]$ for $t > 0, s \in \mathbb{R}$. You are more than welcomed to use Taylor series for simpler integration as you see fit.

Part 4: All About That Velocity Field!

Using the pressure functions you found earlier, deduce the velocity field, knowing that the pressure is inversely proportional to the square of the velocity (by Bernoulli's Principle) by a factor of 2.2 in this case. Then, calculate the circulation of the field, approximating the sinusoidal components as up to their 2nd order Taylor polynomial.

Part 5: What About AoA?

Calculate the field velocity for an AoA of $\alpha = 10^\circ$. Suppose we draw a straight line from the trailing edge to the boundary (the cardioid) and one from the leading edge to the cardioid. The upper region is U and lower region is L . Determine:

$$\Gamma_L = \oint_L \mathbf{v} \cdot d\mathbf{r} \quad \Gamma_U = \oint_U \mathbf{v} \cdot d\mathbf{r}$$